



Optimizing Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety

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Introduction:

Prostate-specific membrane antigen (PSMA) has emerged as a highly promising target for positron emission tomography/computed tomography (PET/CT) imaging in the treatment of prostate cancer. PSMA is overexpressed in prostate cancer cells, enabling precise detection of primary tumors, metastases, and recurrent disease. Radiolabeled PSMA ligands, such as ^{68}Ga -PSMA-11 and ^{18}F -DCFPyL, have demonstrated superior sensitivity and accuracy compared to conventional imaging modalities.

Primary Aim:

To evaluate patient exposure during the PET/CT Prostate procedure.

Methodology:

Study Location:

The investigation was conducted at a PET/CT research center in Almana Hospital, Eastern Province, **Dammam, Saudi Arabia**. This facility serves as one of the country's largest referral hospitals and specialized oncology clinics.

Equipment Specifications:

Patient exposure was assessed using a combined **Siemens Biograph mCT Flow** system, renowned for its precision in nuclear medicine imaging.

Result:

A total of 49 PSMA PET-CT scans were analyzed. The mean age of patients was 68.98 years (SD: 8.76). Patient BMI was distributed across categories, with no single category exceeding 50%. The mean DLP was 921.16 mGy.cm (SD: 253). The mean CTDI was 7.66 mGy (SD: 1.54). The mean radionuclide activity was 265 MBq (SD: 12).

Discussion:

Our study's mean activity (265.12 MBq) was notably lower than Fendler et al. (269 MBq) and substantially reduced compared to Salah et al. (314.4 MBq).

This reduction suggests successful dose optimization while maintaining diagnostic quality, supporting ALARA principles in clinical practice.

Evidence supports the feasibility of reduced-activity protocols without compromising diagnostic accuracy in PSMA PET/CT imaging.

Conclusion:

The findings underscore the need to optimize imaging protocols to balance diagnostic accuracy with patient safety. Future research should focus on dose-reduction strategies, such as advanced reconstruction techniques or lower-dose CT protocols, to minimize radiation exposure while maintaining high clinical efficacy in the management of prostate cancer.

References:

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Table 1: Demographic Data

Statistic	Age (Y)	Height (Cm)	Wight (Kg)	BMI
Mean	68.98	170.54	80.37	27.65
StDiv	8.76	7.45	15.37	5.15
Min	42	154	54	19.72
Max	89	184	131	44.28

Table 2: Dose Parameters

Statistic	Activity (MBq)	DLP (mGy.cm)	CTDI (mGy)	Scan Length (mm)	SSDE (mGy)	KVp
Mean	265.12	921.16	7.66	1167.55	9.66	120.41
StDiv	12.12	253.71	1.54	291.30	1.64	2.86
Min	236.8	485.54	5.93	314	7.09	120
Max	296	2211.82	16.92	1707	17.43	140

Table 3: Compare our study with the previous study

Author	Activity (MBq)	
Current Study	265.12	Saudi Arabia
(Fendler et al., 2023)	269	European
(Salah et al., 2021)	314.4	Saudi Arabia